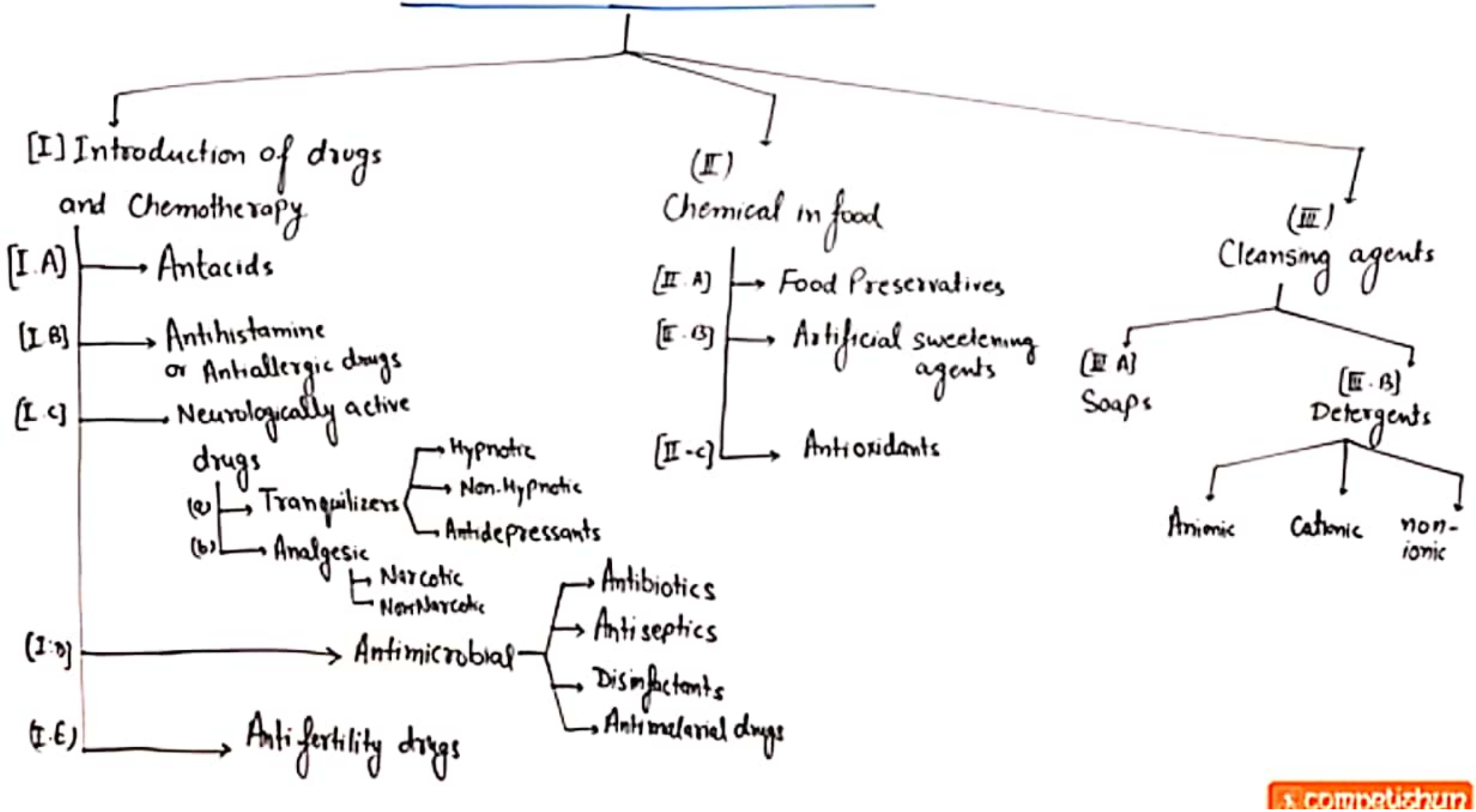
CHEMISTRY IN EVERYDAY LIFE



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CHEMISTRY IN EVERYDAY LIFE

(I) Introduction of drugs and Chemotherapy:-

⇒ **Drugs:-** Drugs are low molecular mass chemical which interact with macromolecule targets and produce a biological respose.

⇒ **Medicines:-** Medicines are chemicals that are usful in diagnosis, prevention and treatment of diseases.

⇒ **Therapeutic effect:-** Desirable and beneficial effect of drug, like treatment of symptoms and cure of disease on a living body is known as therpeutic effect.

⇒ **Chemotherapy:-** The use of chemicals to distroy infectious micro organisms without causing any injury to the host is called chemotherapy.

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CHEMISTRY IN EVERYDAY LIFE

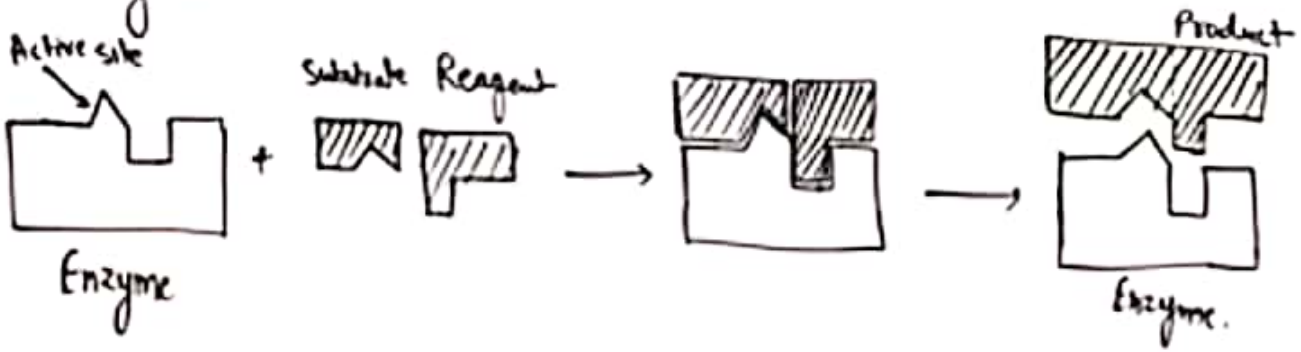
(I) Introduction of drugs and Chemotherapy:-

⇒ **Enzymes:-** Proteins which preform the role of biological catalyst in the body

⇒ **Function of enzymes:-**

(i) Active sites of enzymes hold the substrate molecule in a suitable position, so that it can be attacked by reagents effectively

(ii) Enzyme can provide a functional group that will attack the substrate and carry out reaction



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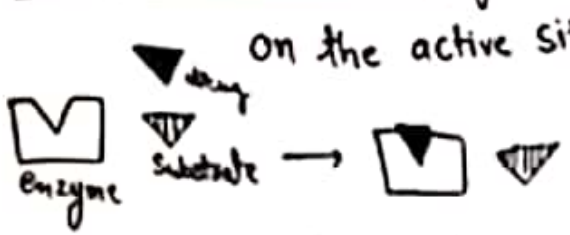
(I) Introduction of drugs and Chemotherapy:-

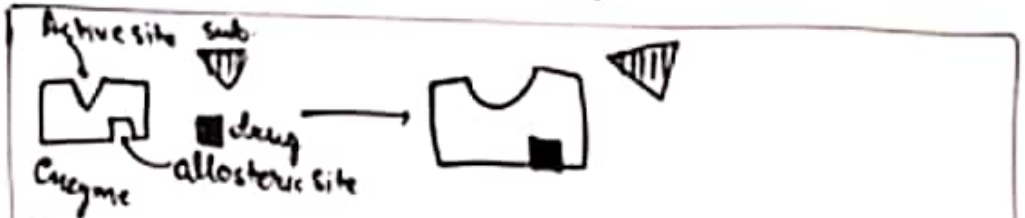
⇒ **Role of drugs:-** Either increase or decrease role of enzyme catalysed reaction.

⇒ Inhibition of enzymes is common role of drug action

⇒ **Enzyme inhibitor:-** Enzyme inhibitors is drug which inhibits catalytic activity of enzymes or blocks the active sites of the enzyme eventually prevents the binding of substrate with enzyme.

→ (i) **Competitive inhibitor drugs:-** That compete with the natural substrate for their attachment on the active sites of enzymes.





→ (ii) **Non competitive inhibitor drugs:-** Those drugs bind to a allosteric site and changes the shape of the active site in such a way that substrate can not recognise.

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(I) Introduction of drugs and Chemotherapy:-

Types of drugs:-

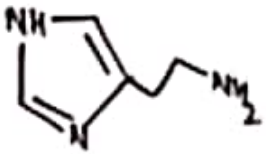
[I A] **Antacids:-** Chemical substances which neutralize excess acid in gastric juices and give relief from acid indigetion, acidity, heart burns & gastric ulcers.

Examples → Eno, gelusil, digene, baking soda

→ $Al(OH)_3$, $Mg(OH)_3$

→ Cemitidine, Ramtidine, lansoprazol, Omeprazole etc.
(Tegamet) (zantac)

⇒ **Histamine:-** It stimulates the secretion of pepsin and HCl in stomach



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CHEMISTRY IN EVERYDAY LIFE

(I) Introduction of drugs and Chemotherapy:-

Types of drugs:-

[I.B] **Antihistamines:-** Chemical which diminish or abolish the effects of histamine released in body and hence prevent allergic reactions.

- Histamine is vasodilator, It has various functions. It contracts the smooth muscles in bronchi and relax

Examples: ✓Brompheniramine (Dimetapp)
✓Terfenadine (seldane)
Chlorpheniramine (zeet)
✓Diphenyl hydramine (Benadryl)
Pheniramine maleate (avil)

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CHEMISTRY IN EVERYDAY LIFE

(I) Introduction of drugs and Chemotherapy:-

Types of drugs:-

[I C] Neurologically active drugs
- ① Tranquilizers
- ② Analgesics.

└ Drugs which have a neurological effect

② Analgesics:- Chemical used to relieve pain without causing any disturbances in the nerous system.

ⓐ Non-narcotic (Non addictive) used as antipyretics.

Prostaglandins

Example: Aspirin, Paracetamol, Ibuprofen, Dichlofenac Sodium etc.

ⓑ Narcotic :-

Morphine, Heroin, Codeine

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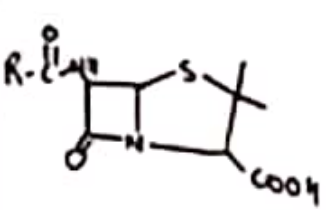
(I) Introduction of drugs and Chemotherapy:-

(I D) Antimicrobials:-

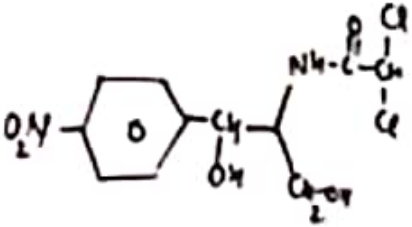
Drugs that tends to destroy/prevent development on inhibit the pathogenic action of microbes such as bacteria (antibacterial drug), fungi (antifungal drugs) virus (antiviral agents) or other parasites (anti. parasitic drugs) selectively.

① **Antibiotics:-** The chemicals produced by micro organism like bacteria, fungi, moulds that inhibit the growth or destroy other micro organism causing infectious disease in men or animal's body, called antibiotics.

Bactericidal	Bacteriostatic
Penicillin	Erythromycin
Aminoglycoside	Tetracycline
ofloxacin	Chloramphenicol



General structure of penicillin



Chloramphenicol

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CHEMISTRY IN EVERYDAY LIFE

(I) Introduction of drugs and Chemotherapy:-

(ID) Antimicrobials:-

① Antibiotics:-

⇒ Penicillin:-

Ampicillin and, amoxycilline are used for the treatment of pneumonia.

⇒ Streptomycin is used for the treatment of tuberculosis, meningitis & pneumonia.

⇒ Tetracyclins:-

Teramycine, Oriomycin

└ for typhoid └ for eyes

⇒ Chloramphenicol:- for typhoid, meningitis and pneumonia.

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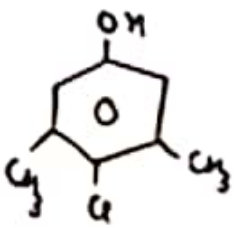
CHEMISTRY IN EVERYDAY LIFE

(I) Introduction of drugs and Chemotherapy:-

(ID) Antimicrobials:-

② Antiseptic:- Chemical substances that kill or prevent growth of microorganisms and can be applied on living tissues such as cuts, wounds and diseased skin surface are called antiseptics.

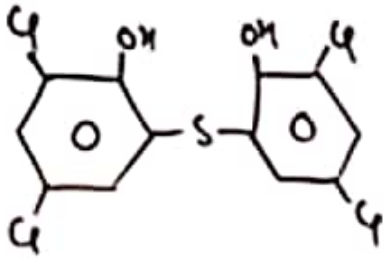
Example:- Furacin, Sofromycin, chloroxylenol, Terpineol, Bithional, iodoform, Tincture of iodine, 0.2% solution of phenol.



Chloroxylenol



Terpineol



Bithional

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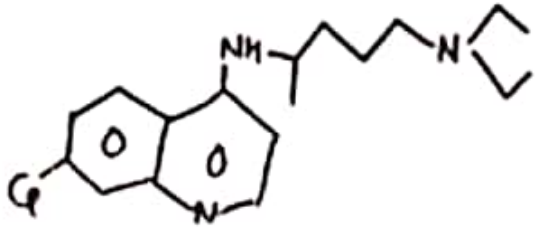
(I) Introduction of drugs and Chemotherapy:-

(I D) Antimicrobials:-

(3) **Disinfectants:-** Chemical substances that kill microorganisms but can not be applied on living tissues.

examples =
- 1% solution of phenol
- 0.2 to 0.4 ppm Cl_2 solution is used to disinfect drinking water
- Hexachlorophen
- Thymol
- methylene blue

(4) **Antimalarials:-** Chloroquine



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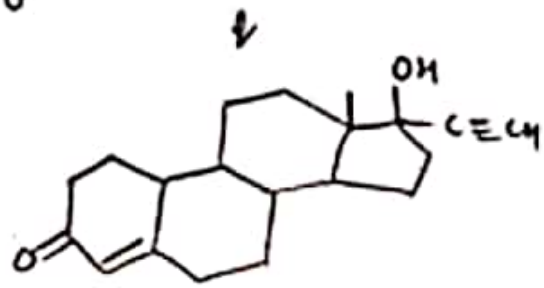
(E) Introduction of drugs and Chemotherapy:-

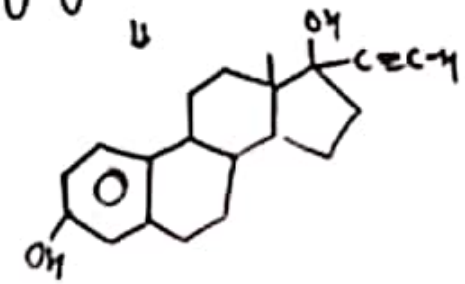
(I E) Antifertility drugs:-

Chemical substances which are used to control pregnancy in women are called anti-fertility drugs or birth control pills or oral contraceptives.

these drugs are synthetic derivatives of estrogen and progesteron

eg Norethindron , Ethynylestradiol (Novestrol)





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CHEMISTRY IN EVERYDAY LIFE

(II) Chemical in Food :-

Chemicals which are added to food for their presarration, enhancing their appearance, taste, odour and nutritive value are called food additives.

(II.A) Food preservatives :- Chemicals which are added to food material to prevent their soilage due to microbial growth, yeast and molds.

Examples :- ① Sodium benzoate (0.06% to 0.1%) - used for preservation of fruit juice, Jam, Jelly, Pickles etc.

② Parabens = (Alkyl-p-hydroxy benzoate) - " " " tomato sauce, soft drinks, beer & Syrups

③ Sorbates = (Salt of sorbic acid) - " " " milk cheese, meat & fish.

④ Propionates = (ethyl / phenyl propionate) - " " " biscuits and baked products.

⑤ Epoxides :- " " " nuts, dried fruits.

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CHEMISTRY IN EVERYDAY LIFE

(II) Chemical in Food :-

(II.B) Artificial Sweetening agents :-

Chemical compounds which gives sweetening effect to the food and enhance its flavour.

- These are highly valuable for diabetic persons and people who need to control intake calories.
- use in pan masala, ice cream, drinks, mouthwash, toothpaste, toffies etc.

Examples
1. Saccharine (550)
2. Aspartame (100) → Unstable at cooking temp, (in cold drinks) / Methyl ester of dipeptide.
3. Sucralose (600) → Trichloro derivative of sucrose.
4. Alitame (2000)

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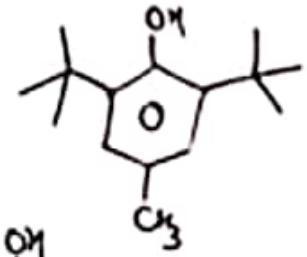
(II) Chemical in Food :-

(II-C) Antioxidant :-

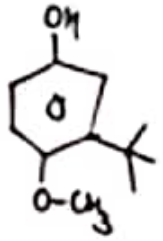
Chemical Substance which reduce the rate of oxidation of food material.

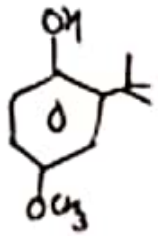
Examples

① BHT = 2,6-Ditertiary butyl hydroxyl toluene



② BHA = Butylated hydroxy anisol





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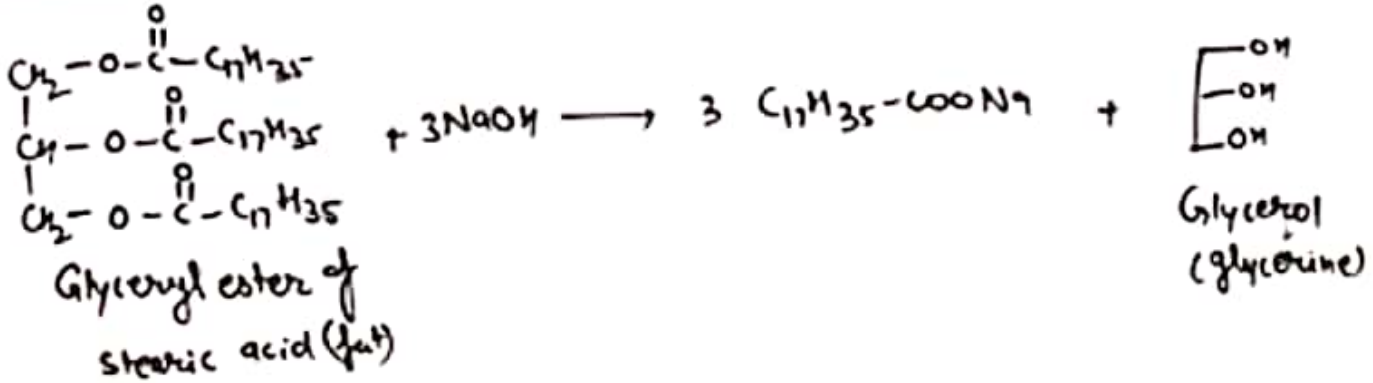
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CHEMISTRY IN EVERYDAY LIFE

(III) Cleansing agents:-

(A) Soaps:- Soaps are Na/K salts of long chain fatty acids like Palmitic acid ($C_{15}H_{31}-COOH$), Stearic acid ($C_{17}H_{35}-COOH$) Oleic acid ($C_{17}H_{33}-COOH$), linoleic acid ($C_{17}H_{31}-COOH$) etc.

Saponification:-



Examples

(1) Toilet Soap
(2) Transparent soaps
(3) Medicated soap
(4) Shaving Soaps
(5) Laundry soaps.
etc.

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CHEMISTRY IN EVERYDAY LIFE

(III) Cleansing agents:-

(A) Soap:-



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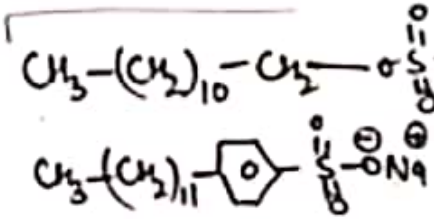
CHEMISTRY IN EVERYDAY LIFE

(III) Cleansing agents:-

(B) Detergents:- The synthetic products, which like soaps remove dust and grease from a surface are called detergent.

detegent are mainly sodium salts of sulphuric/ sulphonic acid with long chain hydrocarbons

(a) anionic detergents:-

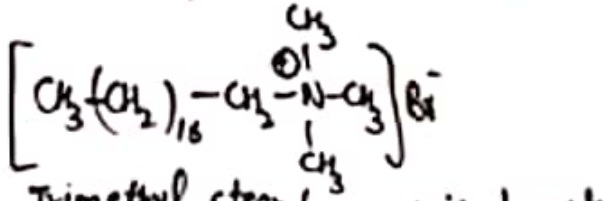


(Sodium Lauryl sulphate)

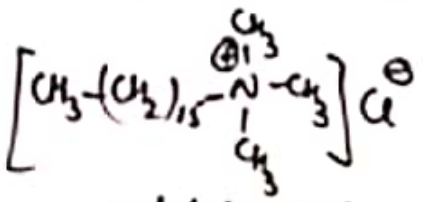
(Sod. dodecyl benzene sulphonate)

(b) Cationic detergent:- Cl^-, Br^-, CH_3COO^- salts of quaternary amine, having long hydrocarbon alkyl group.

- ⇒ germicidal
- ⇒ expensive



Trimethyl stearyl ammonium bromide



Cetyl trimethyl ammonium chloride

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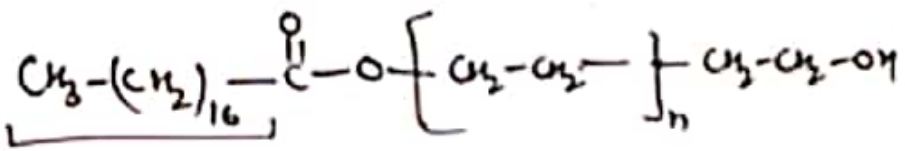
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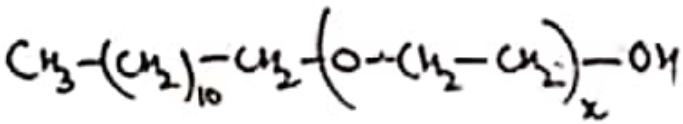
(III) Cleansing agents:-

(B) Detergents:-

(c) **Non ionic detergents:-** Esters of glycol of high molecular mass.



Poly ethylene glycol stearate



Lauryl alcohol ethoxylate

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